

Example: Compare the average daily particulate-matter concentration 64/20 M ①
for Zone A, zone B. (in $\mu\text{g}/\text{m}^3$)

Assume both population are normally distributed.

$$\bar{x}_1 = 43.0 \quad \bar{x}_2 = 39.8$$

$$n_1 = 15 \quad n_2 = 20$$

$$\sigma_1 = 4.1 \quad \sigma_2 = 3.6$$

a) Test $H_0: \mu_1 = \mu_2$ vs. $H_a: \mu_1 > \mu_2$.

$$H_0: \mu_1 - \mu_2 = 0 \quad \text{vs.} \quad H_a: \mu_1 - \mu_2 > 0$$

$$\alpha = 0.01$$

1) Test statistic.

$$Z^* = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{(43.0 - 39.8) - 0}{\sqrt{\frac{4.1^2}{15} + \frac{3.6^2}{20}}} = 2.41$$

2) critical value.

$$Z_\alpha = Z_{0.01} = 2.33$$

3) Decision and conclusion.

$$Z^* = 2.41 > Z_\alpha = 2.33$$

Therefore, reject H_0 .

We have strong enough evidence to support $H_a: \mu_1 > \mu_2$.

b) p-value = $P(Z > Z^*)$

$$= P(Z > 2.41)$$

$$= 1 - P(Z \leq 2.41) = 1 - 0.9920 = \boxed{0.008}$$



②

$$p\text{-value} = 0.008 < 0.01 = \alpha.$$

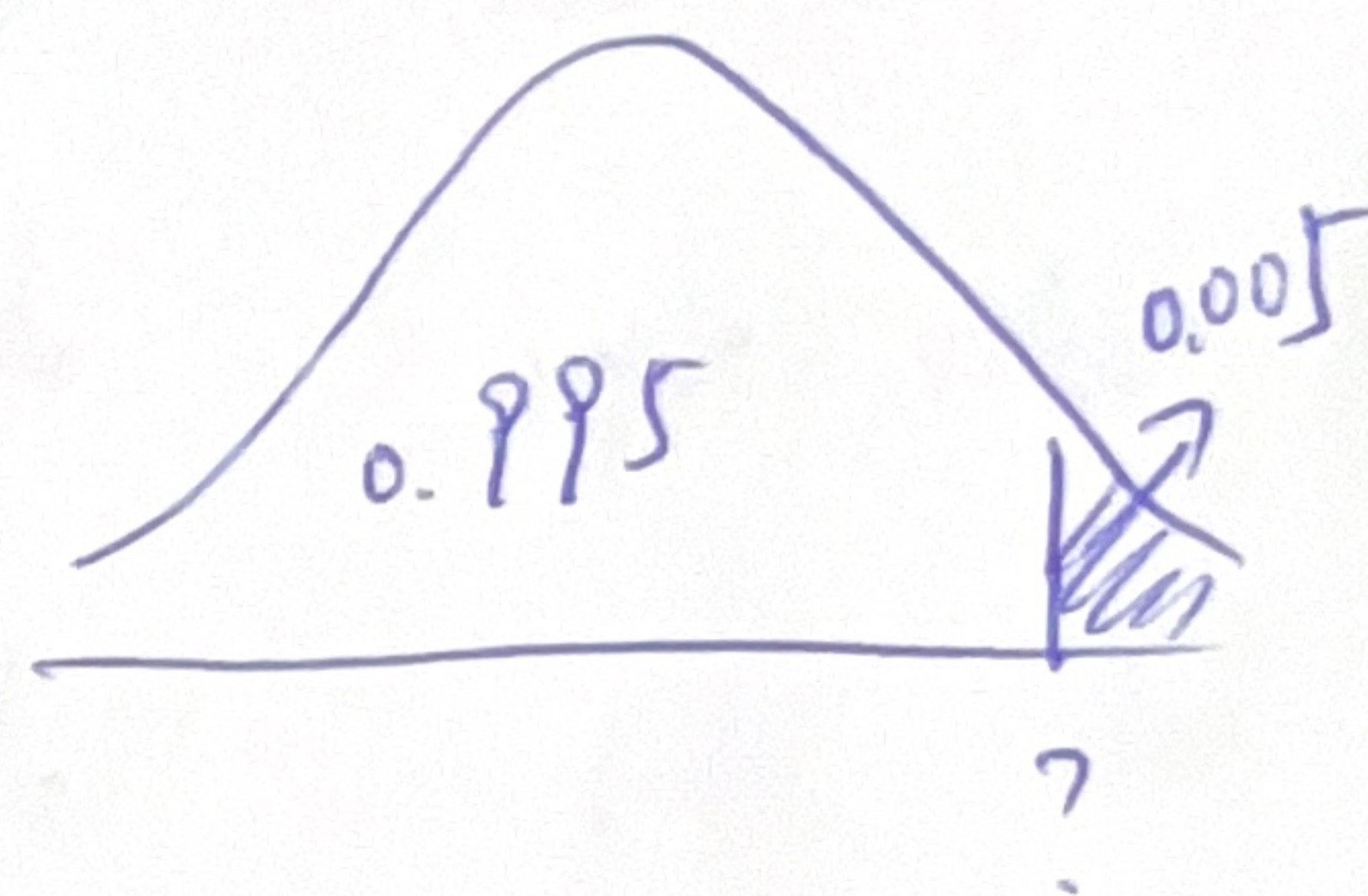
Therefore, reject H_0 .

c) 99% two-sided CI for $(\mu_1 - \mu_2)$

$$(\bar{x}_1 - \bar{x}_2) \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$= (43.0 - 39.8) \pm \underbrace{Z_{0.005}}_{2.575} \sqrt{\frac{4.1^2}{15} + \frac{3.6^2}{20}}$$

$$= (-0.2245, 6.6245)$$



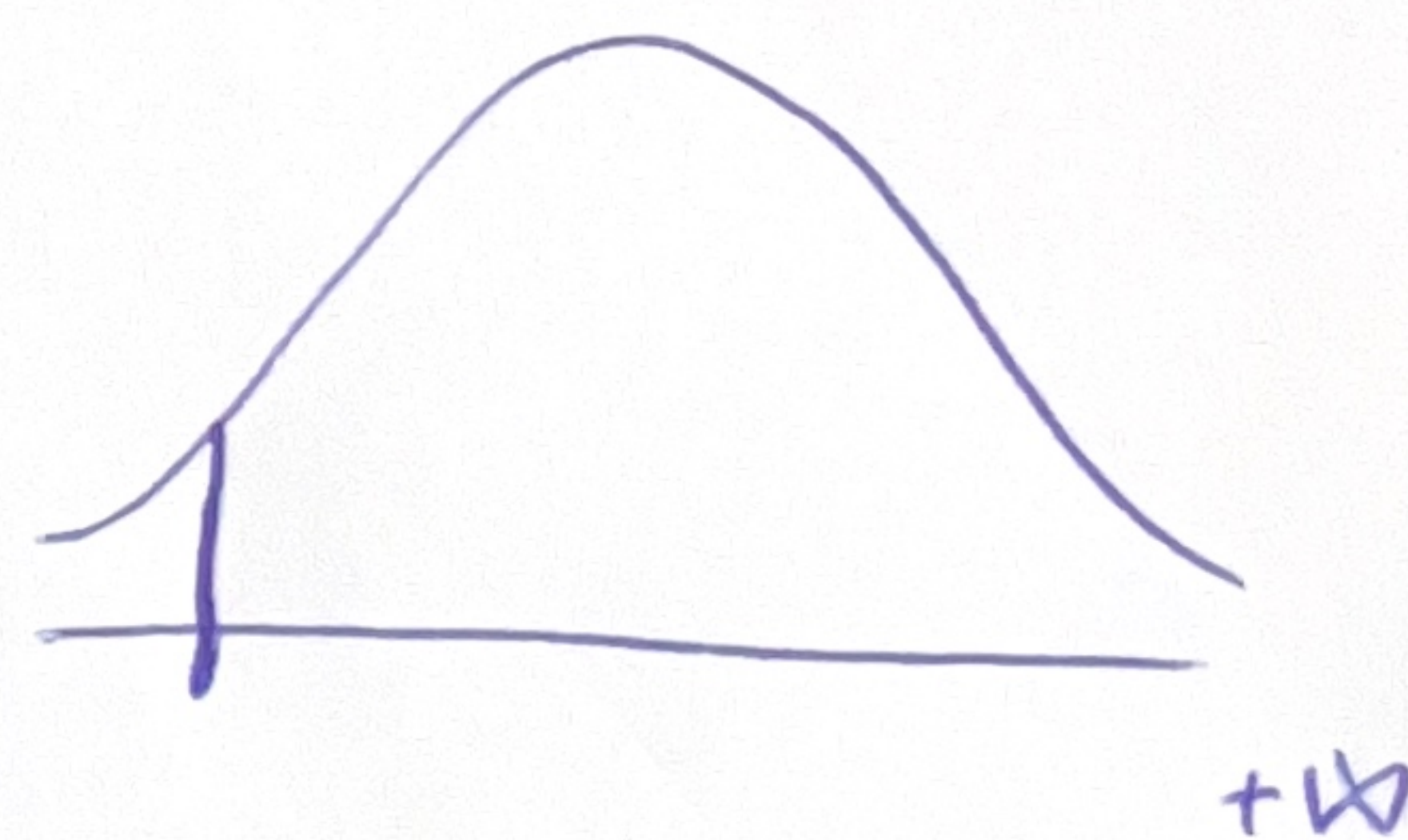
d) Is $\mu_1 = \mu_2$ a reasonable guess? based on c)?

Yes. 0 is in CI.

e) 99% lower-bound CI for $\mu_1 - \mu_2$.

$$\left((\bar{x}_1 - \bar{x}_2) - Z_{\alpha} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}, +\infty \right)$$

$$= (0.10, +\infty)$$



f) Is $\mu_1 = \mu_2$ a reasonable guess based e)?

No. 0 is not in CI.

(3)

Hypothesis test for $\mu_1 - \mu_2$ but σ_1, σ_2 unknown

Assuming $\sigma_1 = \sigma_2$.

~~3-step test.~~

$$H_0: \mu_1 - \mu_2 = d_0 \quad \text{vs.} \quad \begin{cases} H_a: \mu_1 - \mu_2 \neq d_0 & \text{two-sided} \\ H_a: \mu_1 - \mu_2 > d_0 & \text{right-sided} \\ H_a: \mu_1 - \mu_2 < d_0 & \text{left-sided} \end{cases}$$

3-step test:

▷ Test statistic.

$$t^* = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\text{where } s_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}$$

2> critical value.

two-sided: $t_{\frac{\alpha}{2}, n_1 + n_2 - 2}$.

right-sided: $t_{\alpha, n_1 + n_2 - 2}$.

left-sided: $-t_{\alpha, n_1 + n_2 - 2}$.

3> Decision and conclusion:

two-sided: If $|t^*| > t_{\frac{\alpha}{2}, n_1 + n_2 - 2}$, then reject H_0 .

We have strong enough evidence to suggest H_a .

right-sided: If $t^* > t_{\alpha, n_1 + n_2 - 2}$, then reject H_0 .

left-sided: If $t^* < -t_{\alpha, n_1 + n_2 - 2}$, then reject H_0 .

4) p-value :

two-sided : $p\text{-value} = 2 P (t_{n_1+n_2-2} > |t^*|)$

right-sided : $p\text{-value} = P (t_{n_1+n_2-2} > t^*)$

left-sided : $p\text{-value} = P (t_{n_1+n_2-2} < t^*)$

If $p\text{-value} < \alpha$, then reject H_0

5) $(1 - \alpha) 100\%$ CI. for $\mu_1 - \mu_2$.

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\frac{\alpha}{2}, n_1+n_2-2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$